

Messages for Roli

In Memory of Deepak Kapur

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ThanhVu (Vu) Nguyen

Former PhD Student of Deepak's · Professor, George Mason University

It was a great shock when Roli told me about Deepak's passing on Saturday. We were in the middle of an NSF proposal submission, and Deepak had just sent me suggestions on Wednesday. Despite his deteriorating health, he was still very active in research and eager to learn new things, and his thinking was very sharp in his last days as it was when he was in his prime.

In the beginning of my PhD study at UNM, I was working on program repair using genetic programming, and we were quite successful in fixing real bugs. However, Deepak was skeptical about the lack of formal specification. He kept asking about the meaning of "correct" and "bug", which I could not provide, and this led him to question the work: "how can you fix what you don't understand?" Deepak transitioned me to his "rigorous" camp, and my PhD thesis is about dynamic invariant generation and more generally formal program analysis, which forms the basis of all my work since then.

Despite being a respected senior researcher, Deepak could be quite amusing and fun to be around. I always called him "Professor" (just Professor and not Professor Kapur) from the very beginning of my PhD. He found that amusing and started to call me "Professor" as well, which is how we addressed each other until I actually became a professor, at which point he started using my name. He also liked free food, and when he saw announcements from UNM events with food, he would send me to go and get some for him. I remember that once the Daily Lobo had an ad for a restaurant offering a heavy discount — \$1 for a taco/sandwich — and I grabbed 20-something newspapers from the Daily Lobo office, and we went there for lunch for almost a month. Clearly mischievous, but also a lot of fun.

When my first paper with him received the Distinguished Paper award at ICSE — the top conference in software engineering — he said "that tells him a lot about the quality of SE research." But that is how he was: brutally honest and never mincing his words, which of course could easily offend people. But if people could get past that, as I and many others who knew him well can attest, they would find that Deepak cared a lot for his colleagues, students, and especially about CS education and research — perhaps a bit too much, such that he got frustrated when he saw "junk" research or students lacking foundational knowledge. I should note that many CS faculty today feel exactly

the same way about the state of education in the age of AI and LLMs. The difference is that most of us are not as blunt as Deepak and therefore have much better RateMyProfessor ratings.

Deepak is the type of person that I simply do not believe could “retire” and would think about research until his last day. I have been working on neural network verification (NNV) in recent years, and Deepak was quite interested in this line of research and tried to learn about it — at the age of 73, knowing he had just a couple of years left! In fact, just a few weeks ago, I was describing to him my new paper on generating proofs to certify results of NNV tools, and he immediately observed that the resulting proofs can be used to modify the original networks and make them much smaller. I was quite excited about this idea and decided to invite him to be a co-PI on an NSF proposal I am submitting. His last message to me (Wed 4/8/2026 8:57PM) was about research. I imagine that in those last few days, he was still thinking about how to make NNV proof checking more efficient.

Shuang (Sean) Luan, PhD

Professor and Chair, Department of Computer Science, University of New Mexico

Deepak had a profound impact on UNM Computer Science. He joined the department in 1999 as Chair and helped transform it from primarily a teaching department into a research active one. To support faculty research, he significantly increased the number of TA lines, providing essential teaching support and serving as an important recruiting tool for top graduate students. These changes also helped reduce faculty teaching loads, allowing more time for research.

I joined in 2004 and directly benefited from these efforts. As Chair, Deepak recruited Darko Stefanovic (former long-time department chair), Jared Saia, Patrick Bridges (CARC director), Melanie Moses (too many roles to list from the Dean’s Office to the Provost Office and the state), and myself, all of whom are now full professors and form the backbone of the department today.

Deepak truly loved both research and teaching. He cared deeply about his students and was dedicated to supporting their growth, both academically and personally. In a way, it may be a blessing that he passed while doing what he loved throughout his life.

Deepak’s office and mine are just one door apart, and we were both usually early. For years, after dropping off my kids at school, I would walk down the empty hallway in Farris to my office. I would see his door ajar. I would say, “Good morning, Deepak!” and he would reply, “Good morning, Sir!” — even though he was far more senior than I am, he always responded that way.

We would often chat over coffee, and sometimes a snack, in the mornings, sharing stories. I know he hated Starbucks Caffè Verona. He was always curious about whatever new sweet treat I had brought in. He would accept them, though reluctantly, while still insisting that he really shouldn’t. He will be missed dearly by everyone in the department.

Gruia-Catalin Roman

Former Dean, University of New Mexico · Close Friend

I lost a dear friend and it is painful. He meant so much to me, both as a respected colleague and as a special human being. We first met when I came to UNM as Dean. Deepak was suspicious of my “brilliant” and “bold” initiatives based on a lifelong distrust of authority, well founded in his personal experiences. Later, we had offices close by and a deep friendship evolved. We shared similar values, we loved to analyze world affairs, we enjoyed retelling the story of our lives, and we encouraged each other to face challenges straight on. I loved our dinners together, especially those involving Roli’s wonderful cooking, and I appreciated his collection of great Scotches, which he generously offered for side-by-side tasting. He enjoyed movies and often recommended hard to watch traumatic stories about life in India.

Deepak loved both teaching and research. I think he was a perfectionist in both. I had the chance to see him teach and many a times we discussed pedagogy and the challenges students are facing. He showed care for the individual while demanding quality work; he accepted no shortcuts on his part and invested much effort to ensure an excellent learning experience, even when facing major health challenges. On the research side, he felt a deep obligation toward his graduate students and their future careers. He was a great mentor and rejected retirement out of a desire to see his last student graduate. Professionally, he was an excellent researcher who continued to maintain research funding until the very end. His dedication to intellectual inquiry and innovation were central to his identity and he could not envision a life without research.

I learned a lot from Deepak and will always think of him with affection. His passing is a great loss for our academic institution, his friends, and his family. Sarita and I love Roli and we are by her side in these difficult times.

S N Maheshwari (Sachin)

Professor, Department of Computer Science and Engineering, IIT Delhi

Sanjiva just told me about the passing away of Deepak. I am sad more than shocked. Every year when we met he seemed to have physically aged. But his conduct both physical and mental was so sharp that focus was never on the internal changes that were perhaps taking place. His involvements continued to be what they always had been. No letting off on any grounds. I met him first in the early 80s at the annual FST TCS conferences. What stood out was that unlike many others he

always submitted his better papers at FST TCS. This was particularly impressive as there were not many among those who attended then with whom he could exchange ideas and/or collaborate. But he was always there. Looking back I would say he helped lay the foundation of automated theorem proving in India.

Over the last fifteen or twenty years my memories of Deepak are more of a social nature — time spent together on the Delhi campus on your annual visits. To me it was important that his visits to IIT be as smooth as they could be. Greatness had to be looked after particularly when the person wore it lightly.

I am sad for many reasons. I will miss his company, little discussions over sweet chai, and his aura. But your loss is much more profound. Only thing I will say to you is that rare persons like Deepak do not really go gently into night. They leave too much behind, not only in our memories but in the collective living of the community where his ideas will continue to challenge and inspire future generations for a long time.

Thomas Wies

Professor, Department of Computer Science, NYU · Deepak's Collaborator During Sabbatical

Please accept my deepest condolences, also from my partner Ruzica Piskac, who knew Deepak for many years. We have been very saddened by this news.

Deepak was visiting me last summer and fall when he spent his sabbatical at NYU. Deepak told me not long ago that he and Roli bought an apartment in New York and that they were looking forward to spending more time in the city. He and I had plans to continue our collaboration once he had time to fully settle in New York.

I first met Deepak at a Dagstuhl seminar many years ago. Since I was attending the seminar locally from Saarbruecken, I was there by car and gave him a ride back at the end of the week. The car was fully packed with PhD students and Deepak did not mind at all. He was so kind and generous to us all. I always admired him — not only was he an exceptional scientist but also such a truly wonderful and kind person. It is a great loss. We will dearly miss him.
